

DEEP LEARNING · PROJECT PROPOSAL

Waste Image Classification Using Deep Learning

A proposed CNN-based approach to recognising six waste categories from images and supporting better segregation.

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PROJECT CONTEXT

1–3. Abstract and Introduction

Abstract

This project proposes a convolutional neural network (CNN) that will classify waste photographs into Plastic, Paper, Glass, Metal, Cardboard and Trash. The intended system will accept an uploaded image, prepare it for the model and return a predicted category. It aims to support waste management through accessible image-based guidance.

The planned work includes dataset preparation, model development, training, evaluation and a web interface. Training and deployment have not been confirmed, and this report makes no performance claims.

Waste management and segregation

Waste management includes collecting, sorting, recycling and disposing of waste. Classification is important because correctly separated materials can enter more suitable recycling streams. Mixing different materials or contaminated items can reduce recycling quality and increase disposal needs.

Why use deep learning?

Manual segregation is slow, inconsistent and can expose workers to unsafe materials. People may also confuse objects that look similar but belong to different categories.

Deep learning can learn visual patterns directly from labelled images. A CNN is suitable for this proposed task because it can learn features such as edges, textures and shapes without manually defining every recognition rule.

> **Intended use**
>
> The classifier will provide sorting guidance. A predicted material label will not replace local recycling rules or safe handling procedures.

RESEARCH SCOPE

4–5. Problem Statement and Objectives

Problem statement

Waste sorting depends on recognising materials correctly. Similar-looking packaging, mixed waste and inconsistent judgement can lead to incorrect segregation. Manual sorting is also labour-intensive and error-prone.

The proposed system will recognise a likely waste category from a photograph and present it through an upload interface. Its usefulness will need to be established through evaluation on unseen images.

Project objectives

1 **Build an image-based classifier**

Accept waste photographs as the model input.

2 **Identify waste categories**

Assign images to the six defined class labels.

3 **Apply CNNs**

Learn visual features for material recognition.

4 **Train and evaluate the model**

Measure performance using held-out data.

5 **Provide an upload interface with predictions**

Display the predicted class and confidence score.

6 **Support smarter segregation and recycling**

Make material guidance easier to access.

6. Dataset Description

A labelled Garbage Classification image dataset is planned, organised into one folder per class. The photographs below are AI-generated category illustrations, not verified dataset samples.



Plastic



Paper



Glass



Metal



Cardboard



Trash

CLASS	DESCRIPTION	IMAGE COUNT
Plastic	Plastic bottles and packaging.	To be updated after actual model training and evaluation.
Paper	Sheets and paper products.	
Glass	Glass bottles and jars.	
Metal	Cans and metal items.	
Cardboard	Corrugated boxes and board.	
Trash	Residual waste under dataset labels.	

Dataset source, total images, image dimensions and class distribution: To be updated after actual model training and evaluation.

Images will be resized and normalised. Training, validation and test sets will be separated; split ratios remain to be confirmed.

7. Proposed System Architecture

The proposed pipeline will turn an uploaded photograph into one predicted waste category. Feature extraction and classification are parts of the CNN, shown separately here to explain their roles.

1

Input Waste Image

The user will supply a photograph containing the waste item to be classified.

2

Image Preprocessing

The image will be resized and its pixel values normalised to match the model input.

3

CNN Model

The prepared image will pass through the network using learned model weights.

4

Feature Extraction

Convolutional layers will learn visual patterns such as edges, textures and shapes.

5

Classification

The classification layers will convert the learned features into scores for the six categories.

6

Predicted Waste Category

The category with the highest output probability will be displayed with its confidence score.

Sequential workflow · Input → Preprocessing → CNN → Features → Classification → Category

8. CNN Model Architecture

A CNN is proposed for six-class image classification. Layer counts, filter sizes and other implementation settings will be finalised during model development.

Input

A resized, normalised image enters the network.

↓ Convolutional layers

Learn filters that detect edges, textures and more complex visual patterns.

↓ ReLU

Adds non-linearity by replacing negative activations with zero.

↓ Max Pooling

Reduces feature-map size while retaining strong responses.

↓ Flatten

Converts feature maps into a single feature vector.

↓ Dense

Combines the extracted features to produce class scores.

↓ Softmax

Produces a probability for each of the six classes.

Conceptual flowchart · Convolution, ReLU and pooling may be repeated. Final architecture is pending.

9. Methodology and Implementation

The following implementation sequence is planned. Dataset decisions and evaluation records will be documented before deployment is considered.

Planned methodology

1. Dataset collection
2. Dataset organisation
3. Image preprocessing
4. Dataset splitting
5. CNN model development
6. Model training
7. Validation
8. Evaluation
9. Saving trained weights
10. Image prediction
11. Web application deployment

Implementation controls

Training data will be used to learn model weights. Validation data will guide model choices, while the test set will be reserved for final evaluation.

The same resizing, normalisation and class-label mapping must be used during training and image prediction.

Saved weights will be linked to the final model definition so that inference can reproduce the intended pipeline.

Proposed tools

TOOL	ROLE
Python 3.12	Programming and workflow coordination.
TensorFlow/Keras	CNN development, training and inference.
NumPy	Numerical arrays and image operations.
Pandas	Dataset records and result summaries.
Matplotlib	Training and evaluation plots.
Seaborn	Confusion-matrix visualisation.
Scikit-learn	Data splitting and evaluation metrics.
Pillow	Image loading and resizing.

10–11. Training, Evaluation and Results

Training and evaluation are pending. Actual performance discussion will be added after the model is tested; no results are claimed.

ITEM	MEANING	VALUE
Epochs	Complete passes through training data.	To be updated after actual model training and evaluation.
Batch size	Images processed per weight update.	
Loss function	Measures prediction error.	
Optimizer	Adjusts weights to reduce error.	
Accuracy	Proportion classified correctly.	
Validation loss	Error on validation images.	
Confusion matrix	Counts actual versus predicted labels.	
Precision	How often a predicted class is correct.	
Recall	How many true class examples are found.	
F1-score	Balances precision and recall.	

Reserved evidence spaces

Prediction screenshots

Pending actual predictions

Accuracy graph

Pending training history

Loss graph

Pending training history

Confusion matrix

Pending test evaluation

12–13. Web Application and Benefits

The proposed web application will provide a simple route from a waste photograph to a predicted category. The interface and model-serving arrangement have not yet been deployed or confirmed.



Illustrative image capture, not an application screenshot.

Proposed user flow

1. Upload a waste image.
2. Resize and normalise it.
3. Run model prediction.
4. Show the predicted class and confidence score.

Confidence will represent model output, not a guarantee that the classification is correct.

Planned delivery

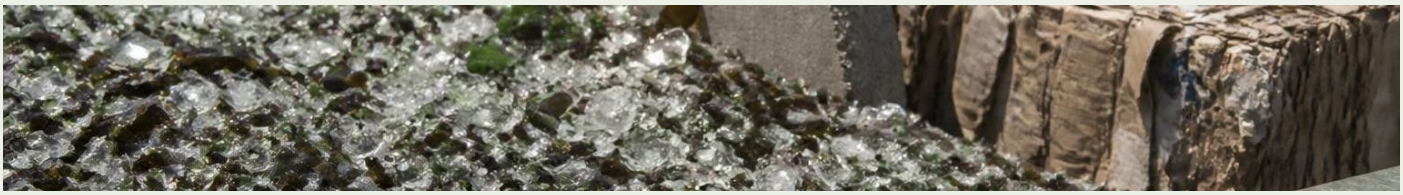
GitHub is planned for source code and model weights, subject to repository and file-size requirements. Vercel is planned for the web interface. The inference hosting approach and deployment compatibility still require verification.

Expected benefits

- Support automated segregation through image-based guidance.
- Help users identify materials for recycling.
- Encourage environmental awareness.
- Reduce repetitive manual identification effort.
- Provide a foundation for future smart waste management systems.

These are intended benefits, not measured outcomes.

14–16. Limitations, Future Work, Conclusion and References



Material separation remains the practical goal.

Limitations

Similar-looking materials, poor lighting and limited dataset diversity may reduce recognition quality. Class imbalance can bias predictions, while mixed or contaminated waste may not fit a single material label.

Future work

Future development could use larger and more diverse datasets, compare stronger model architectures and investigate real-time camera detection. A mobile application and integration with smart recycling bins are possible extensions after reliable evaluation.

Conclusion

This project proposes a CNN-based waste image classifier and an upload interface to support cleaner recycling. Its practical value remains to be tested. Dataset verification, model training, independent evaluation and deployment checks are required before effectiveness can be claimed.

References

- TensorFlow documentation. **tensorflow.org**.
- Keras documentation. **keras.io**.
- LeCun, Y., Bottou, L., Bengio, Y. and Haffner, P. “Gradient-Based Learning Applied to Document Recognition.” *Proceedings of the IEEE*, 1998.
- Dataset source: To be updated after actual model training and evaluation. Add the confirmed source and citation.

Project status / Proposed system; training results and deployment confirmation pending.